



Earth Fault Indicator

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In mains-powered equipment, exposed metal parts are connected to earth wire in order to prevent users from contact with high voltages if electrical insulation fails. Connections to ground through earth connection also limit the build-up of static electricity when handling electrostatic-sensitive devices. Earth in a mains electrical wiring system is a conductor that provides a low-impedance path to the earth to prevent hazardous voltages from appearing on equipment and hence the name.

Here is a simple tester to find whether the mains wiring is correct or not. This tester can be used to check sockets periodically before connecting appliances like heater and electric iron. It indicates the status of the mains wiring through two LEDs as shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit of earth fault indicator. The circuit is built around a BC547 transistor (T1), two LEDs (LED1 and LED2), two 1N4007 diodes (D1 and D2) and five resistors (R1 through R5).

The circuit takes advantage of the voltage that appears across the earth and neutral terminals. Neutral-to-earth

voltage as measured at the load for a single-phase circuit is a function of the load current and the impedance of the neutral wire. Various standards limit this voltage drop in a branch circuit to 3 per cent (5 per cent total for feeder and branch circuit) for a reasonable efficiency of operation. Based on this, the neutral-to-earth voltage limit for a single-phase 120V AC circuit is 3.6V AC and for a single-phase 230V AC circuit 6.6V AC.

There is no additional power supply used to operate this circuit. The circuit is directly powered from the 230V AC mains supply. The combination of diode D1 and resistor R1 reduces the 230V AC mains to a low voltage for the circuit. A transistor switch is provided

to light up the green LED (LED1) if earth is correctly connected. The base of T1 is connected to earth pin of the mains supply through a network of resistors R2 and R3 as shown in Fig. 2.

Working of the circuit is simple. The red LED (LED2) lights up if there is power in the socket and phase (L) and neutral (N) lines are connected correctly. Diode D2 protects the green LED (LED1) from damage when the polarity changes. Resistors R4 and R5 limit current through LED1 and LED2, respectively.

When the mains wiring is proper, a potential difference develops between the neutral (N) and earth (E) lines and transistor T1 turns on to light up the green LED (LED1). This indicates that the earth connection is perfect. At the same time, the red LED also glows, indicating that the phase (L) and neutral (N) lines are connected properly. In brief, if the phase, neutral and earth connections are proper, both the red and green LEDs light up. When the earth connection is broken, the red LED2 glows but green LED1 does not.



Fig. 1: Author's prototype

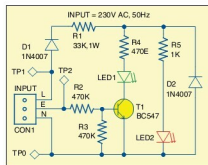


Fig. 2: Circuit of earth fault detector

Test Points

Test point	Details
Across TP0 and TP1	Around 230V AC
Across TP0 and TP2	4V AC (depends on the load current and impedance of neutral wire)

PARTS LIST

Semiconductors:

- LED1 - 5mm green LED
- LED2 - 5mm red LED
- D1, D2 - 1N4007 rectifier diode
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, 5% carbon, unless stated otherwise):

- R1 - 33-kilo-ohm, 1W
- R2, R3 - 470-kilo-ohm
- R4 - 470-ohm
- R5 - 1-kilo-ohm

Miscellaneous:

- CON1 - 3-pin connector for mains

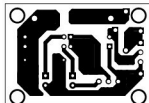


Fig. 3: An actual-size, single-side PCB for the earth fault indicator

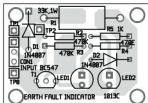


Fig. 4: Component layout for the PCB

Construction and testing

An actual-size, single-side PCB for the earth fault indicator is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable plastic case. The circuit can be housed in an adaptor box with three pins (see Fig. 1).

To test the circuit for proper functioning, check input supply at TP1 with respect to TP0. Also check the voltage difference across neutral and earth pins as per the test point table.

EFY note. As the circuit is operated directly from the mains voltage, it should be assembled only by experienced persons. To avoid lethal shock, do not touch or troubleshoot the circuit when it is connected to mains power. ●

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